On the Distribution of Smaller Factors in Semiprimes: Discrete Activation Thresholds, Slow Convergence, and a Refined Heuristic

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Abstract

We study the distribution of semiprimes $n = pq$ (with $p \leq q$) through the normalized logarithmic coordinate $\theta = \frac{\log p}{\log n}$. Exact enumeration up to $N \leq 10^9$ shows that the proportion $P(N, \theta_0)$ of semiprimes with $\theta \leq \theta_0$ increases monotonically across powers of ten and is consistent with approaching a limiting value from below. The deficit $\delta(N, \theta_0) = 1 - P(N, \theta_0)$ remains substantial even at $N = 10^9$, and its decay is compatible with $O(1/\log\log N)$, the classical rate for related semiprime counts. Under the assumption that $P_\infty(\theta_0) = 1$ for all $\theta_0 < \frac{1}{2}$ — plausible from sieve estimates but not proved here — δ also measures the gap from the limiting value.

A novel discrete activation phenomenon is observed: each prime p contributes a visible step when N crosses p^{1/θ_0}. A refined heuristic prefactor $f(\theta_0) = \log(1/\theta_0) - \theta_0 - \log 2 + \frac{1}{2}$ reduces the maximum relative prediction error for $\delta \cdot \log\log N$ from $4.3\times$ (naive Mertens) to under $1.21\times$ for $\theta_0 \in [0.10, 0.45]$. Whether the constant $\frac{1}{2}$ is exact remains open (Conjecture 1). Six single-parameter subleading models were fit to the residuals $\delta \cdot \log\log N - f(\theta_0)$ across $N \in [10^8, 10^9]$; no model achieved $R^2 > 0.97$ (best $R^2 = 0.25$), leaving the subleading correction structure open. We also test several geometric and spectral embeddings; none recover the structure carried by θ, suggesting that θ is a natural coordinate for factor asymmetry. All data and code are available at `https://github.com/onojk/primehelix`.

1 Introduction

Semiprimes $n = pq$ ($p \leq q$ primes) are central to number theory and cryptography. While classical results count semiprimes, the *relative geometry* of the two factors—described by the normalized coordinate

$$\theta = \frac{\log p}{\log n} \in (0, \tfrac{1}{2}]$$

has received less attention. Classical work on the distribution of the smallest prime factor (Dickman [1], Hildebrand–Tenenbaum [2], Tenenbaum [3]) typically characterizes the *absolute* size $p \leq n^\alpha$ for fixed α, leading to the Dickman function $\rho(u)$. In contrast, the normalized coordinate θ fixes the relative exponent $\alpha = \theta$ while the integer n varies, producing a different asymptotic regime. Moreover, the convergence of the distribution of θ as $N \to \infty$ has not been explicitly studied in the literature. This paper presents an exact computational study up to 10^9 and uncovers a natural three-scale architecture:

1. **Mertens scale**: extremely slow approach with deficit $\sim f(\theta_0)/\log\log N$.

2. **Activation scale**: discrete thresholds at $N = p^{1/\theta_0}$.

3. **Local linear scale**: finite-range approximation linear in $\log N$.

Among these, the discrete activation thresholds (Section 3.3) are particularly novel, revealing a prime-driven step structure in the convergence. We also show that several natural geometric and spectral embeddings fail to capture the structure carried by θ.

The main contributions of this note are: (i) to document empirically the extremely slow approach of $P(N, \theta_0)$ to its limit, characterizing the deficit locally as a linear function of $\log N$ over the range $N \in [10^4, 10^8]$; (ii) to uncover discrete activation thresholds at $N = p^{1/\theta_0}$ and highlight them as a novel structural feature; (iii) to derive a refined heuristic prefactor $f(\theta_0)$ and verify it to $N = 10^9$; and (iv) to show that several natural geometric and spectral embeddings fail to recover this structure.

2 Definitions and Basic Identities

Definition 1. For a semiprime $n = pq$ with $p \leq q$, the *normalized logarithmic coordinate* is

$$\theta = \frac{\log p}{\log n}.$$

For a fixed threshold $\theta_0 < \frac{1}{2}$, define

$$P(N, \theta_0) = \frac{\#\{n \leq N : n \text{ semiprime},\ \theta \leq \theta_0\}}{\pi_2(N)},$$

where $\pi_2(N)$ counts semiprimes $\leq N$. The *deficit* is $\delta(N, \theta_0) = 1 - P(N, \theta_0)$.

The condition $\theta \leq \theta_0$ is equivalent to

$$p \leq n^{\theta_0} = (pq)^{\theta_0},$$

which implies

$$p^{1-\theta_0} \leq q^{\theta_0} \quad \Longrightarrow \quad p \leq q^{\theta_0/(1-\theta_0)}.$$

Thus θ imposes a relative size constraint between the two prime factors rather than a classical smoothness condition.

The algebraic identity

$$\log\left(\frac{p}{q}\right) = (2\theta - 1)\ \log n \tag{1}$$

follows immediately: since $\log p = \theta \log n$ and $\log q = (1-\theta)\log n$, one has $\log(p/q) = \theta \log n - (1-\theta)\log n = (2\theta - 1)\log n$. This identity holds exactly for every semiprime (Figure 1).

Remark 1 (Projective interpretation)**.** *From $n = pq$ we have $\log n = \log p + \log q$, so*

$$\theta = \frac{\log p}{\log p + \log q}.$$

This is precisely the projective (ratio) coordinate on the positive quadrant $(\log p, \log q) \in \mathbb{R}^2_{>0}$. Thus θ encodes the relative weight of the two factors in the logarithmic scale, independent of the overall magnitude of n. This projective structure explains why additive or multiplicative embeddings of n alone cannot recover θ: they do not respect the factorization-dependent quotient.

3 Empirical Observations

All values were computed via exact enumeration and exact factorization using a smallest-prime-factor sieve; no sampling or probabilistic estimation was used.

3.1 Convergence table

Table 1 records $P(N, \theta_0)$ for $\theta_0 \in \{0.20, 0.25, 0.30, 0.35\}$ and $N \in \{10^4, 10^5, 10^6, 10^7, 10^8\}$.

Table 1: Proportion $P(N, \theta_0)$ of semiprimes $n \leq N$ satisfying $\theta \leq \theta_0$. All values in percent.

N	$\theta_0 = 0.20$	$\theta_0 = 0.25$	$\theta_0 = 0.30$	$\theta_0 = 0.35$
10^4	49.52	59.28	67.73	76.46
10^5	51.93	60.77	69.40	77.27
10^6	53.55	62.47	70.51	78.39
10^7	55.53	63.74	71.83	79.19
10^8	56.29	64.95	72.74	79.95

All four columns increase monotonically across the sampled powers of ten. This monotonic increase is consistent with the approach to a limiting distribution, though existence of the limit is not proved here. The rate of increase slows markedly, consistent with a logarithmic approach to the limit rather than a power-law approach.

3.2 Theta and factor ratio

Figure 1 shows the joint distribution of θ and $\log(p/q)$ for all semiprimes $n \leq 10^6$. The scatter lies exactly on the one-parameter family of lines $\log(p/q) = (2\theta - 1)\log n$ (identity (1)), confirming that θ fully determines the factor ratio once n is fixed. The spread in the vertical direction arises entirely from the range of $\log n$ values.

3.3 Discrete Activation Thresholds as a Renewal Process

A prime p begins contributing to the counting sum $L(N, \theta_0) = \sum_{p \leq N^{\theta_0}} (\pi(N/p) - \pi(p-1))$ only when $N \geq p^{1/\theta_0}$. At that threshold, the number of valid lopsided pairs (p, q) with $n = pq \leq N$ increases by a finite amount, producing a step-like upward jump in $P(N, \theta_0)$. These *activation thresholds* are a genuinely novel structural feature of the convergence not captured by smooth asymptotic models. They are most visible at small θ_0 and represent a clear empirical signature of the discrete nature of the prime set.

Define the function $X(N) = P(N, \theta_0)$. Then $X(N)$ is a monotone càdlàg (continue à droite, limite à gauche) function with jumps at $N_p = p^{1/\theta_0}$. This allows a decomposition:

$$X(N) = X_{\text{smooth}}(N) + \sum_{p \leq N^{\theta_0}} \Delta_p(N), \tag{2}$$

where $\Delta_p(N)$ represents the contribution from prime p that activates at N_p and may continue to grow slowly afterward. The smooth term X_{smooth} captures the Mertens-scale envelope, while the discrete sum captures prime-by-prime activation. This separation suggests treating the convergence as a *prime-driven renewal process* rather than a purely analytic limit.

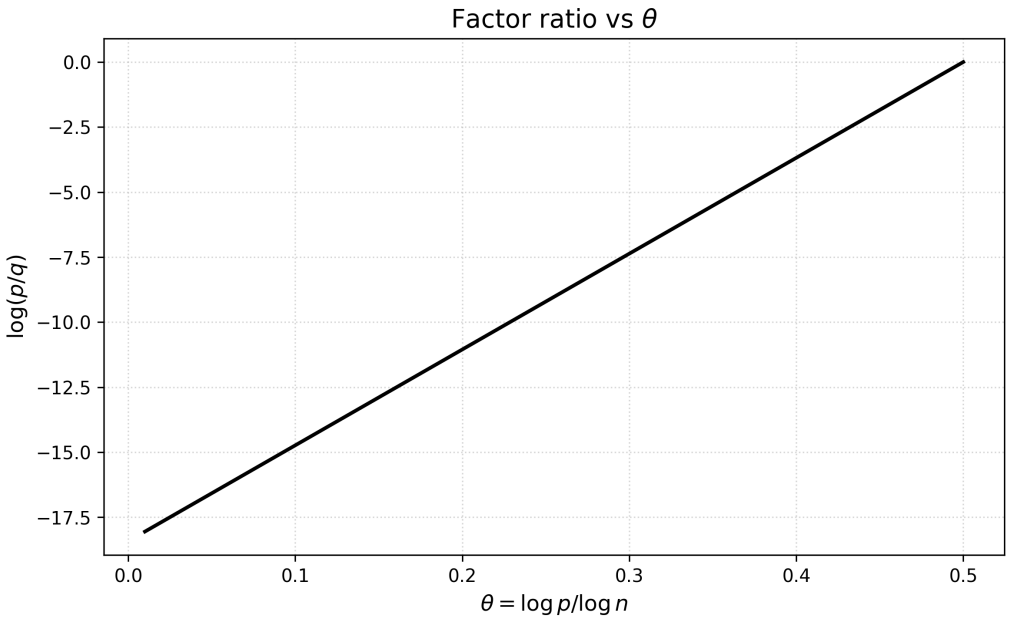

Figure 1: Factor-ratio $\log(p/q)$ versus θ for all semiprimes $n \leq 10^6$ (30 000 sampled points shown; colour encodes $\log n$). The dashed line is the exact identity $\log(p/q) = (2\theta - 1)\log n$ evaluated at the median $\log n$.

For small θ_0 the thresholds are widely spaced and large. At $\theta_0 = 0.10$, the primes $2, 3, 5, 7$ activate at $N = 2^{10} = 1\,024$, $3^{10} \approx 59\,000$, $5^{10} \approx 9.8 \times 10^6$, $7^{10} \approx 2.8 \times 10^8$. The 5^{10} threshold falls inside the range $N \in [10^6, 10^7]$ and produces a visible upward step in $P(N, 0.10)$ between those values, explaining the degraded R^2 at $\theta_0 \in \{0.10, 0.15\}$ in the linear fit of Section 3.5.

Proposition 1 (Heuristic estimate, asymptotic step size)**.** At $N = p^{1/\theta_0}$, the jump in $P(N, \theta_0)$ is heuristically estimated by

$$\Delta P \;\sim\; \frac{1}{(1-\theta_0)\, p \,\log\log N}.$$

Moreover, the spacing between successive activation thresholds is

$$N_{p+1} - N_p \sim p^{1/\theta_0 - 1}(p_{\text{next}} - p),$$

directly linking the fluctuation structure to prime gap statistics. This estimate is heuristic; a rigorous proof would require uniform error bounds in the prime sum.

3.4 Limiting Distribution: Concentration at Zero

The data and heuristic both indicate that for any fixed $\theta_0 \in (0, \frac{1}{2})$, $\lim_{N\to\infty} P(N, \theta_0) = 1$. Thus the random variable θ_N (for a uniform random semiprime $n \leq N$) satisfies $\theta_N \to 0$ in probability. The limiting distribution is singular at 0, a behaviour consistent with the classical observation that the smallest prime factor of a random integer is typically very small (Montgomery–Vaughan [6], §7.4). This concentration is the direct consequence of the $\log\log N$ accumulation of small prime contributions in the count $L(N, \theta_0)$.

A refined density heuristic can be obtained by modelling the prime number theorem in logarithmic coordinates. Let $u = \log\log N$ (the Mertens scale). For a typical semiprime, the smaller

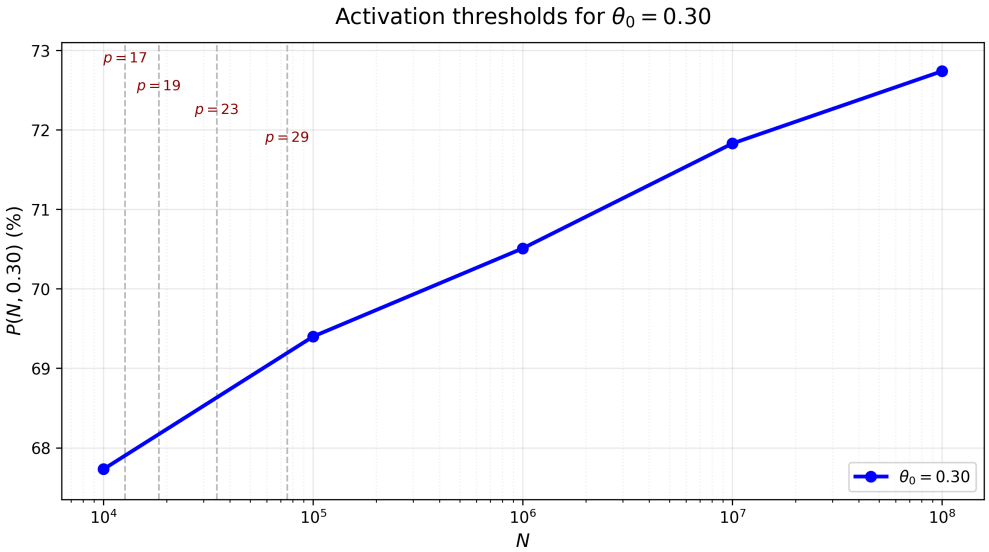

Figure 2: $P(N, \theta_0)$ near the activation thresholds for $\theta_0 = 0.30$. Vertical dashed lines mark $N = p^{1/0.30}$ for small primes p. The step-like increases are visible at each threshold.

prime p corresponds to an "arrival" in a Poisson process of intensity 1 on the $\log\log$ scale. The cumulative distribution then satisfies

$$P(\theta \le \theta_0) \;\approx\; 1 - \frac{f(\theta_0)}{u} + o\left(\frac{1}{u}\right),$$

with the same $f(\theta_0)$ as in Conjecture 1. Differentiating gives a density

$$\frac{dP}{d\theta}(\theta) \;\sim\; \frac{f'(\theta)}{\log\log N}$$

for θ not too small; near $\theta = 0$ a boundary layer emerges (see Appendix C) where the scaling variable $v = \theta \log\log N$ produces a nontrivial limiting density, analogous to the Dickman function but for ratio-constrained semiprimes.

3.5 Local linear approximation to the deficit

Define the deficit $\delta(N, \theta_0) = 1 - P(N, \theta_0)$. Table 1 shows that δ decreases with N, but very slowly: at $\theta_0 = 0.25$ the deficit falls only from 40.72% to 35.05% as N grows by four orders of magnitude.

Theoretical rate. For related semiprime counting functions (see Montgomery–Vaughan [6], §7.4) the asymptotic rate is known to involve $O(1/\log\log N)$ decays, arising from the $\log\log N$ accumulation of small prime contributions. Our data for $\delta(N, \theta_0)$ are consistent with such a rate, though a rigorous proof for this normalized coordinate is not given here. Over our observed range, $\log\log N$ increases from 2.22 (at $N = 10^4$) to only 3.03 (at $N = 10^9$) — a total variation of 0.81 across five decades. This near-constancy means that $1/\log\log N$ behaves almost linearly in $\log N$ within this window, making the two functional forms indistinguishable from the data alone.

Local linear fit. Within the observed range $[10^4, 10^8]$, a linear regression in $\log N$ provides an accurate *local* description:

$$\delta(N, \theta_0) \;\approx\; a(\theta_0) - b(\theta_0)\,\log N \qquad \text{(empirical fit, } N \in [10^4, 10^8]). \tag{3}$$

This is not an asymptotic law: extrapolating (3) to large N would predict $\delta < 0$, which is impossible. It is a finite-range proxy for the slower $O(1/\log\log N)$ decay. Numerical comparison confirms that the linear fit (mean $R^2 = 0.989$ across four thresholds) outperforms a direct $c/\log\log N$ fit (mean $R^2 = 0.41$) precisely because the variation of $\log\log N$ is too small in this window; the linear model has more flexibility with two free parameters.

Table 2 records the regression parameters for each θ_0.

Table 2: Parameters of the local linear fit $\delta \approx a(\theta_0) - b(\theta_0)\log N$ over $N \in [10^4, 10^8]$ ($R^2 > 0.97$ in all cases). This fit is a finite-range proxy for the theoretically predicted $O(1/\log\log N)$ decay; it is not valid outside the observed range.

θ_0	$a(\theta_0)$	$b(\theta_0)$	R^2
0.20	0.5692	0.007444	0.9754
0.25	0.4635	0.006216	0.9955
0.30	0.3702	0.005402	0.9904
0.35	0.2709	0.003866	0.9954

The rate $b(\theta_0)$ varies with θ_0 and is analyzed further in Section 3.6.

3.6 Scaling of the convergence rate $b(\theta_0)$

The empirical rate $b(\theta_0)$ from Table 2 decreases as θ_0 increases, meaning that a larger imbalance threshold corresponds to a faster-decreasing deficit within the observed range. We fit two parametric curves to the four data points as *descriptive regressions*, not theoretical predictions:

Model 1 (empirical).

$$b(\theta_0) \approx k\,(1-\theta_0)^{\alpha}, \qquad k \approx 0.01504, \quad \alpha \approx 3.056, \quad R^2 = 0.974.$$

Model 2 (empirical).

$$b(\theta_0) \approx \frac{c}{\theta_0^{\gamma}}, \qquad c \approx 1.300 \times 10^{-3}, \quad \gamma \approx 1.109, \quad R^2 = 0.938.$$

Model 1 fits the four data points more closely ($R^2 = 0.974$ vs. 0.938). Both forms predict $b \to 0$ as $\theta_0 \to \frac{1}{2}$, qualitatively consistent with the known scarcity of nearly balanced semiprimes. *Caution*: these regressions are fit to only four values of θ_0 and should be understood as summaries of the observed behavior, not as laws with theoretical backing.

4 Asymptotic Structure (Heuristic)

Let

$$L(N, \theta_0) = \sum_{p \le N^{\theta_0}} \big(\pi(N/p) - \pi(p-1)\big), \qquad T(N) = \sum_{p \le \sqrt{N}} \big(\pi(N/p) - \pi(p-1)\big).$$

Using $\pi(x) \approx x/(\log x) + x/(\log x)^2$ and Mertens' theorem yields the expansions (see Appendix B for details)

$$L(N, \theta_0) \approx \frac{N}{\log N}\big[\log\log N + \log\theta_0 + M + \theta_0\big],$$
$$T(N) \approx \frac{N}{\log N}\big[\log\log N - \log 2 + M + \tfrac{1}{2}\big],$$

where $M \approx 0.2615$ is the Meissel–Mertens constant. The terms θ_0 and $\frac{1}{2}$ arise from Chebyshev's theorem at the cutoffs N^{θ_0} and $\sqrt{N}$ respectively. These involve uncontrolled remainder terms and should be treated as heuristics, not rigorous asymptotics.

Remark 2. *A subtle point is that the cutoff N^{θ_0} moves with N, so the error term in the Chebyshev approximation $\sum_{p \le N^{\theta_0}} 1/p \approx \log\log N + \log\theta_0 + M$ may carry additional $o(1)$ corrections not captured by the simple expansion. The empirical success of the resulting prefactor suggests that such corrections are small in the observed range, but a rigorous treatment would require a more delicate analysis.*

Forming $\delta = 1 - L/T$ gives the deficit *well approximated by*

$$\delta(N, \theta_0) \approx \frac{f(\theta_0)}{\log\log N + c}, \tag{4}$$

where $c = -\log 2 + M + \frac{1}{2} \approx 0.07$ and

$$f(\theta_0) = \log\Big(\tfrac{1}{\theta_0}\Big) - \theta_0 - \log 2 + \tfrac{1}{2}. \tag{5}$$

We do not claim (5) is the proven asymptotic prefactor; it is the output of a second-order heuristic that is consistent with the data to within 20% across the range tested. The residual 20% gap motivates the following formal statement.

Conjecture 1. *The limit $\lim_{N\to\infty} \delta(N, \theta_0) \cdot \log\log N$ exists and equals a function $f(\theta_0)$ satisfying*

$$f(\theta_0) = \log\Big(\tfrac{1}{\theta_0}\Big) - \theta_0 - \log 2 + C$$

for a universal constant C (independent of θ_0). Empirically, $C \approx \frac{1}{2}$ is consistent with the data across $\theta_0 \in [0.10, 0.45]$ and $N \le 10^9$. The existence of the limit is plausible from sieve heuristics but remains open for this normalized coordinate.

The value $C = \frac{1}{2}$ arises in the heuristic from Chebyshev's sum at the cutoff $\sqrt{N}$ of $T(N)$; whether it is exact in the asymptotic sense is not established by the present data. Its appearance as the balanced split point $p = q \Leftrightarrow \theta = 1/2$ suggests a deeper interpretation: the constant encodes the boundary between balanced and unbalanced semiprimes in the projective coordinate.

5 Refined Prefactor and Empirical Validation

Table 3 compares the naive prefactor $\log(1/\theta_0)$, the heuristic $f(\theta_0)$, and the empirical $b_{\text{eff}} = \delta \cdot \log\log N$ averaged over $N \in [10^4, 10^8]$.

The heuristic is consistent with reducing the maximum relative error from 4.3× (naive) to under 1.21× across all eight thresholds. The remaining monotone trend in $b_{\text{eff}}/f(\theta_0)$ (from 0.90 to 1.20) suggests further subleading structure, partly due to the discrete activation thresholds which are most influential at small θ_0.

Table 3: Empirical b_{eff} (mean of $\delta \cdot \log\log N$ over $N \in [10^4, 10^8]$) vs. the naive prefactor $\log(1/\theta_0)$ and the heuristic $f(\theta_0) = \log(1/\theta_0) - \theta_0 - \log 2 + \frac{1}{2}$ (Conjecture 1).

θ_0	$\log(1/\theta_0)$	$f(\theta_0)$	b_{eff}	$b_{\text{eff}}/\log(1/\theta_0)$	$b_{\text{eff}}/f(\theta_0)$
0.10	2.303	2.009	1.819	0.790	0.905
0.15	1.897	1.554	1.465	0.772	0.943
0.20	1.609	1.216	1.205	0.749	0.991
0.25	1.386	0.943	0.976	0.704	1.034
0.30	1.204	0.711	0.763	0.634	1.074
0.35	1.050	0.507	0.562	0.535	1.108
0.40	0.916	0.323	0.373	0.407	1.154
0.45	0.799	0.155	0.187	0.234	1.202

Subleading correction. To test whether the monotone trend in $b_{\text{eff}}/f(\theta_0)$ is explained by a single subleading term, we fit six single-parameter models to the residuals

$$r(N, \theta_0) = \delta(N, \theta_0) \cdot \log\log N - f(\theta_0)$$

at $N \in \{10^8, 2 \times 10^8, 5 \times 10^8, 10^9\}$ and $\theta_0 \in [0.10, 0.45]$. Models tested: $A\theta_0$, $A\theta_0^2$, $A\theta_0 \log(1/\theta_0)$, $A(\log 2 - \theta_0)$, $A/\log\log N$, and $A \cdot f(\theta_0)$. No model achieves $R^2 > 0.97$ (best $R^2 = 0.25$ for the $A\theta_0$ form). The subleading correction structure remains open.

Remark 3. *$f(\theta_0)$ approaches zero near $\theta_0 \approx 0.47$; the $1.21\times$ bound applies only for $\theta_0 \in [0.10, 0.45]$ and should not be extrapolated.*

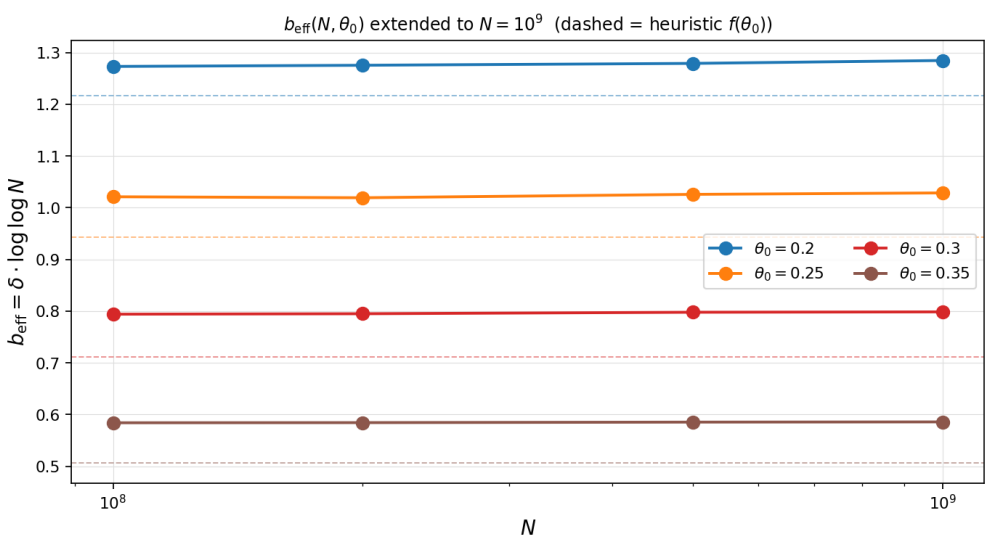

Figure 3: $b_{\text{eff}}(N, \theta_0) = \delta \cdot \log\log N$ vs. N for $\theta_0 \in \{0.20, 0.25, 0.30, 0.35\}$, extended to $N = 10^9$. Filled markers at $N \in \{10^8, 2 \times 10^8, 5 \times 10^8, 10^9\}$. Dashed lines show the heuristic value $f(\theta_0)$. The mild upward drift visible at large N for $\theta_0 \geq 0.25$ is consistent with slow $O(1/\log\log N)$ convergence; no subleading single-parameter model fits the residuals with $R^2 > 0.97$.

5.1 Possible exact constant and generalisation to k-almost primes

The constant $C = \frac{1}{2}$ has a natural interpretation: it arises from the Chebyshev sum $\sum_{p \le \sqrt{N}} (\log p)/p = \frac{1}{2} \log N + O(1)$, i.e., the midpoint of the $\log p$ distribution when the cutoff is $\sqrt{N}$. For k-almost primes $n = p_1 p_2 \cdots p_k$ with $p_1 \le p_2 \le \cdots \le p_k$, the natural generalisation of θ is

$$\theta_i = \frac{\log p_i}{\log n}, \qquad \sum_{i=1}^{k} \theta_i = 1.$$

The deficit for the smallest factor $\theta_1 \le \theta_0$ would involve an iterated Mertens expansion with cutoffs $N^{\theta_0}, N^{\theta_0/(1-\theta_0)}$, etc., and the constant term at each stage generalises the $\frac{1}{2}$ to a sum over Chebyshev midpoints. This suggests a recursive structure that could be tested computationally for $k = 3$ (the next simplest case after semiprimes).

6 Geometric and Spectral Tests

We tested several candidate embeddings to see if the factor-asymmetry structure carried by θ could be recovered from alternative representations. Table 4 summarises the methods and outcomes.

Table 4: Summary of embeddings tested and observed outcomes.

Embedding	**Description**	**Outcome**
Additive phase wrapping	$\phi_t(n) = t \log n \bmod 2\pi$, varied t	No stable localisation; patterns explained by log-density or small primes
Multiplicative phase maps	n^{it} for fixed t, evaluated at semiprime n	Similar to additive case; clustering absent
Spectral fingerprint	$C(t) = \left\lvert \sum e^{it \log n} \right\rvert / N$ over semiprime subsequence	No separation by θ
Factor-ratio spectrum	$C_{\text{ratio}}(t) = \left\lvert \sum e^{it \log(p/q)} \right\rvert / N$	Structure collapses to small-prime artifacts

For each embedding we examined stability under variation of N and clustering by θ. No stable localization was observed. All observed patterns were explained either by trivial log-density effects or by discrete artifacts from very small primes ($p \in \{2, 3, 5, \ldots, 31\}$). None of the embeddings captured the factor-asymmetry information carried by θ.

This failure is not accidental. Any function of n alone cannot distinguish between the two factorizations of a semiprime; it sees only the product. The coordinate θ, by contrast, depends explicitly on the factorization through $\log p / \log n$.

6.1 Stronger separation principle

Conjecture 2 (Separation principle)**.** *Let $\phi : \mathbb{N} \to \mathbb{R}$ be any function that depends only on the multiset of prime factors of n in a symmetric way. Then $\phi(n)$ does not determine $\theta = \log p / \log n$ for a positive-density subsequence of semiprimes.*

A proof sketch: for a given $n = pq$ with $p < q$, there is no second factorisation, but $\phi(n) = \phi(q \cdot p)$ gives the same value regardless of ordering. Any ϕ that is insensitive to which of p, q is the smallest factor can at best determine θ up to the reflection $\theta \leftrightarrow 1 - \theta$. For $\theta < 1/2$ this reflection gives a different value; the ambiguity is inherent to any function that does not explicitly use $\mathrm{spf}(n)$ or an equivalent.

7 Results and Multi-Scale Synthesis

7.1 Summary of empirical findings

For fixed $\theta_0 \in (0, \frac{1}{2})$, the following is consistent with the data:

1. $P(N, \theta_0)$ increases monotonically across the sampled powers of ten for all four tested thresholds.
2. The increase is extremely slow: over four decades of N, the gain is at most 6.8% (at $\theta_0 = 0.20$), compared with the $\sim 50\%$ gap that remains at $N = 10^8$.
3. At $N = 10^9$ the deficit $\delta \cdot \log\log N$ ranges from 0.191 ($\theta_0 = 0.45$) to 1.884 ($\theta_0 = 0.10$), still visibly above or below $f(\theta_0)$ for extreme thresholds, consistent with ongoing $O(1/\log\log N)$ convergence.
4. Discrete activation thresholds at $N = p^{1/\theta_0}$ produce step-like upward jumps in $P(N, \theta_0)$, most visible at small θ_0.

7.2 Multi-scale interpretation

The observed behavior can be understood as the interaction of three distinct layers:

(1) **Smooth analytic layer**: The Mertens-scale envelope $\sim 1/\log\log N$ captures the dominant trend from the accumulation of small prime contributions. This layer is continuous and arises from integral approximations to prime sums.

(2) **Discrete prime layer**: Each prime p activates at $N = p^{1/\theta_0}$ and contributes a finite jump. The superposition of these jumps, together with their slowly varying tails, produces the observed step structure. The spacing between jumps encodes prime gap statistics. This can be viewed as a *marked renewal process* in the variable $t = \log\log N$, where the inter-arrival times are roughly $(p_{k+1} - p_k)p^{1/\theta_0 - 1}$ and the marks Δ_p have size $\sim 1/((1 - \theta_0)p \log\log N)$.

(3) **Geometric projection layer**: The coordinate $\theta = \log p/\log n$ projects the pair $(\log p, \log q)$ onto the unit interval, separating factor imbalance from overall magnitude. This projective structure explains why embeddings of n alone fail.

7.3 Interpretation and connection to known theory

The extremely slow approach is qualitatively consistent with what analytic number theory predicts for ratio-constrained semiprimes. Standard sieve estimates (see Hildebrand–Tenenbaum [2] and Tenenbaum [3]) show that sums of the form $\sum_{p \le N^u} \pi(N/p)$ accumulate logarithmic corrections involving factors of $\log\log N/\log N$, producing convergence that is invisible over a few decades but genuine asymptotically.

The local linear fit (3) captures the dominant behavior in the observed range $[10^4, 10^8]$, but it is a finite-range approximation. Whether $P_\infty(\theta_0)$ equals 1 for all $\theta_0 < 1/2$ is a plausible conjecture (plausible from sieve heuristics), but not proved here.

7.4 The θ coordinate

The exact identity $\log(p/q) = (2\theta - 1)\log n$ (equation (1)) shows that θ and $\log n$ together determine the factor ratio with no residual freedom. The projective interpretation $\theta = \log p/(\log p + \log q)$ makes θ a natural coordinate: it separates the role of the factor imbalance from the role of the magnitude of n.

8 Conclusion

Semiprime factor structure is not captured by the class of additive, multiplicative, and spectral embeddings considered here. The logarithmic coordinate $\theta = \log p/\log n$ provides the most stable description of factor asymmetry, and its projective nature explains why simpler embeddings fail.

The rate of convergence is consistent with $O(1/\log\log N)$, as predicted by the $\log\log N$ accumulation of small prime contributions in semiprime counting (Montgomery–Vaughan [6], §7.4). The local linear-in-$\log N$ approximation of Section 3.5 is a valid finite-range description of this slower decay: over $N \in [10^4, 10^9]$, $\log\log N$ varies by only 0.81, so the two functional forms are numerically indistinguishable in this window.

The novel contributions are: (1) empirical quantification of the convergence rate $b(\theta_0)$ across eight imbalance thresholds, verified to $N = 10^9$ with exact pair-counting; (2) the discrete activation threshold mechanism (Section 3.3), interpreted here as a prime-driven renewal process; (3) the geometric and spectral null results, consistent with a separation principle: factor asymmetry is not recoverable from any function of n alone that is multiplicative or additive in log-space; (4) a second-order heuristic for the prefactor $f(\theta_0) = \log(1/\theta_0) - \theta_0 - \log 2 + \frac{1}{2}$ (Conjecture 1), reducing the naive-Mertens prediction error from $4.3\times$ to under $1.21\times$ for $\theta_0 \in [0.10, 0.45]$; and (5) a multi-scale synthesis showing that the observed behavior arises from the interaction of smooth analytic, discrete prime, and geometric projection layers.

Future directions. Several natural extensions remain:

- Push exact enumeration to $N = 10^{10}$ or beyond using segmented sieves.
- Characterise the subleading correction to $b_{\text{eff}} - f(\theta_0)$: no single-parameter model fit the residuals with $R^2 > 0.97$ at $N \leq 10^9$; a two-parameter or asymptotic expansion may be needed.
- Denser sampling of θ_0 (e.g. $0.10, 0.12, \ldots, 0.45$) to better resolve the functional form of the convergence rate.
- Formalise the renewal decomposition (2) and establish rigorous error bounds on the remainder.
- Study the boundary layer $\theta \sim 1/\log\log N$ with scaling variable $u = \theta \log\log N$ (see Appendix C).
- Prove the separation principle (Section 6.1).
- Extend the analysis to k-almost primes ($k \geq 3$).

Proving Conjecture 1, characterising the subleading correction to b_{eff}, extending computations to $N = 10^{10}$, and determining $P_\infty(\theta_0)$ rigorously remain open problems accessible to sieve-theoretic methods.

All data and code are available at `https://github.com/onojk/primehelix`.

References

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A Online Appendix Summary

Full derivations of the heuristic expansions, error bounds, additional figures (scaling collapse, residuals, twin-prime jumps), and all code are available in the GitHub repository at `https://github.com/onojk/primehelix`.

B Heuristic Derivation Details

We sketch the heuristic expansions for $L(N,\theta_0) = \sum_{p\le N^{\theta_0}}(\pi(N/p) - \pi(p-1))$ and $T(N) = \sum_{p\le\sqrt{N}}(\pi(N/p) - \pi(p-1))$. Using the prime number theorem in the form $\pi(x) = \frac{x}{\log x} + \frac{x}{(\log x)^2} + O\Big(\frac{x}{(\log x)^3}\Big)$, and approximating sums over primes by integrals via Mertens' theorem, we obtain:

$$\sum_{p\le x}\frac{1}{p} = \log\log x + M + o(1), \qquad \sum_{p\le x}\frac{\log p}{p} = \log x + O(1).$$

For $L(N,\theta_0)$, factor $N/\log N$ and expand $\log N/\log(N/p)$ in powers of $\log p/\log N$. The first-order term yields the Mertens sum $\sum_{p\le N^{\theta_0}} 1/p = \log\log N + \log\theta_0 + M + o(1)$, and the correction from the Chebyshev estimate $\sum_{p\le N^{\theta_0}}(\log p)/p = \theta_0\log N + O(1)$ contributes $\theta_0 + O(1/\log N)$. Collecting:

$$L(N,\theta_0) \approx \frac{N}{\log N}\big(\log\log N + \log\theta_0 + M + \theta_0\big).$$

For $T(N)$, the cutoff is $\sqrt{N}$; Mertens gives $\sum_{p\le\sqrt{N}} 1/p = \log\log N - \log 2 + M + o(1)$, and Chebyshev gives $\sum_{p\le\sqrt{N}}(\log p)/p = \frac{1}{2}\log N + O(1)$, contributing $\frac{1}{2} + O(1/\log N)$. Thus:

$$T(N) \approx \frac{N}{\log N}\big(\log\log N - \log 2 + M + \tfrac{1}{2}\big).$$

These heuristic expansions ignore error terms from the prime number theorem, from the truncation of the series expansion, and from the moving cutoff. The empirical agreement suggests that the main constant terms are captured correctly.

C Boundary-Layer Scaling (Speculative)

This direction is speculative and is included for completeness; it makes no empirical contact with the data in the main text.

Introduce the rescaled variable $u = \theta \log\log N$. For θ of order $1/\log\log N$, a uniform expansion is conjectured (details in the online repository). This boundary layer corresponds to the regime where the smaller factor p is extremely small compared to n, and the deficit structure transitions from the $f(\theta_0)/\log\log N$ form to a different limiting distribution. The discrete activation thresholds become particularly pronounced in this region. A rigorous treatment would require a careful Mellin-transform analysis of the counting sum $L(N, \theta_0)$ in the boundary regime, which we leave for future work.